

Activation Energy and Catalysts

Energy profiles, temperature, the Arrhenius equation and catalysts

Name _____ Class _____ Date _____

USEFUL RELATIONSHIPS

$$k = A e^{-E_a/RT} \cdot \ln \frac{k_2}{k_1} = \frac{E_a}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right) \quad R = 8.314 \text{ J/(mol}\cdot\text{K)} \cdot 1 \text{ kJ} = 1000 \text{ J} \cdot \text{T in kelvin}$$

Convert E_a to J/mol before substituting, and show units in every step.

A. Understanding

1. Define activation energy and transition state.

2. A reaction has E_a (forward) = 120 kJ/mol and $\Delta H = -40$ kJ/mol. (a) Is it exothermic or endothermic? (b) Find E_a for the reverse reaction. (c) A catalyst lowers the forward E_a to 70 kJ/mol. What is the reverse E_a now, and what is ΔH ?

3. Use the Maxwell-Boltzmann distribution to explain why a small increase in temperature can roughly double the rate of a reaction.

4. Explain the difference between a homogeneous and a heterogeneous catalyst, with one example of each, and state the effect of a catalyst on K .

B. The Arrhenius equation

5. A reaction has $E_a = 75.0 \text{ kJ/mol}$. By what factor does k increase between 300. K and 310. K?

6. A rate constant is $3.0 \times 10^{-5} \text{ s}^{-1}$ at 500. K and $6.0 \times 10^{-4} \text{ s}^{-1}$ at 550. K. Calculate E_a in kJ/mol.

7. For a reaction with $E_a = 75.0 \text{ kJ/mol}$, $k = 2.0 \times 10^{-4} \text{ s}^{-1}$ at 300. K. Calculate k at 330. K.

C. Challenge

8. A catalyst lowers E_a from 75.0 kJ/mol to 50.0 kJ/mol at 298 K. Assuming the frequency factor A is unchanged, by what factor does the rate constant increase?

9. An Arrhenius plot of $\ln k$ against $1/T$ is a straight line with slope $-6.65 \times 10^3 \text{ K}$. Calculate E_a .